

Structural Labels for Stratified Audit Budget Allocation in Knowledge-Graph Dependency Flows

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Abstract

Knowledge-graph and retrieval-augmented information systems expose dependency flows, entity bindings, and verification records that must be audited under fixed budgets. A scalar score does not indicate which artifacts are bottlenecks, which evidence is redundant, or what downstream records become visible. We systematize fixed-budget audit allocation around graph-topology labels and evaluate a label-guided Life-Saving First policy with Impact Coverage@K. Countries-KG serves as a clean implementation fixture. A complementary controlled knowledge-maintenance experiment uses a corrected multidisciplinary Wikidata substrate with 2000 nodes and 6039 edges. At the prespecified 25% budget, Flat Top-K attains higher reachability coverage than Life-Saving First (0.993 versus 0.953), whereas Life-Saving First protects more uniquely at-risk records (0.972 versus 0.895). At 5%, its small coverage advantage over Flat Top-K does not confirm across 16 discipline anchor clusters after Holm correction ($p = \mathbf{0.385}$). Edge deletion and insertion further degrade the stratified policy more than Flat Top-K on clean-reference coverage. The findings support the audit-record representation and expose a budget-dependent coverage-protection trade-off; they do not establish a universally superior allocation policy, native Wikidata roles, production effectiveness, human usefulness, or causal identification.

Keywords: knowledge graphs, structural labels, audit budget allocation, impact coverage, information systems, dependency flows

1 Introduction

Knowledge-intensive information systems increasingly expose intermediate records rather than only final answers. A knowledge-graph workflow may expose entity nodes, relation paths, retrieval passages, verification steps, and dependency links. These artifacts are useful for maintenance only when a curator can decide what to inspect under a fixed budget. The difficulty is not simply that some artifacts receive low scalar scores. A curator also needs to know whether an artifact is a bottleneck for later records, whether another artifact provides redundant coverage, and which downstream nodes would be affected if the selected artifact is audited.

This paper therefore moves the contribution away from a scoring race and toward audit-record representation rather than score improvement. It provides a systematization of fixed-budget audit allocation through structural label extraction and label-guided budget allocation. We propose a *structural label extractor*, not a scorer, that turns directed graph topology into audit-ready boolean fields:

bottleneck and **redundancy**. These fields are not reward-model predictions and are not claimed to improve task reasoning. They are reusable labels for audit records in dependency-flow settings.

The second contribution is a *stratified budget allocation* policy, Life-Saving First. The policy first allocates budget to Critical Bottleneck nodes, then to Unique Evidence nodes, then to Redundancy Group Samples, and finally to Fallback nodes. Raw risk score is used exclusively as a tie-breaker within the same stratification layer, not as the primary allocation driver. Flat Top-K uses the shared raw risk score as the sole ranking criterion, where the KG setting defines it as trace-local min-max normalized downstream impact count.

The evaluation follows this reframing. Countries-KG is retained as a 30-entity, 189-triple implementation fixture. The main external substrate is a 2000-node Wikidata scientist-institution-field subgraph with controlled maintenance motifs, fixed-budget comparisons, edge perturbations, anchor-cluster confirmation, efficiency measurements, and verified revision-history examples. The Wikidata experiment evaluates controlled audit scenarios on a real KG substrate rather than native Wikidata structural roles.

Figure 1 gives the high-level knowledge-engineering workflow. Structural label extraction sits between raw dependency-flow artifacts and fixed-budget audit allocation: it represents artifacts as directed dependency graphs, converts topology into reusable labels, and allocates inspection budget by structural strata.



Fig. 1 Controlled knowledge-maintenance workflow. A real Wikidata substrate is extracted and layered before controlled audit motifs are added; structural labels guide fixed-budget selection, Impact Coverage measures downstream visibility, and revision-history cases provide maintenance context without implying historical deployment.

2 Related Work

2.1 Attribution and Explanation

Attribution methods ask which parts of an input, computation, or structure matter for a decision. Integrated Gradients [1], SHAP [2], and LIME [3] return feature-level or local explanation signals, while ERASER evaluates extractive rationales [4].

Structure-aware attribution adds graph context. Network studies show that importance can be non-local, redundant, or concentrated around bottleneck nodes [5, 6]; GNNExplainer selects compact explanatory subgraphs [7]. Human-centered XAI further shows that explanations must fit review tasks and organizational routines [8–11].

These lines motivate the audit-triage setting. Saliency identifies important evidence; audit triage additionally needs role labels for local support, duplicated evidence, downstream dependencies, and weak upstream signals. Entailment-tree and counterfactual methods add intermediate or contrastive evidence [12, 13]; the present framework keeps the audit roles separate.

2.2 Process Supervision and Process Reward Models

Process supervision assigns annotation fields to intermediate artifacts as well as final answers. PRM800K showed that per-artifact human labels can support verifier-guided best-of- N search relative to outcome reward models [14]; GSM8K made this verifier setting concrete [15]. Process-based feedback also reduced reasoning errors when outcome success appeared saturated [16]. These studies provide the artifact-resolved fidelity field that the present framework uses as input.

A parallel line makes verification, critique, and revision available as process units. Chain-of-thought and self-consistency expose sampled reasoning paths [17, 18]. Tree of Thoughts and ReAct add

search or action structure [19, 20], while Reflexion and Self-Refine turn revision steps into supervision targets [21, 22].

Recent process-verification resources also show that step-level signals need distributional checks. Math-Shepherd studies automatic step verification without human annotations [23], and Process-Bench evaluates process-error identification across models [24]. These resources help assess scalar step signals, which SC-FMA uses as inputs to structural calibration.

2.3 Graph-Based Knowledge Engineering and Audit Methods

Graph-based knowledge engineering shows how structured intermediate states can be made inspectable. Knowledge-engineering methods have long treated representation, inference, and maintenance-oriented review as coupled design problems [25]. Early expert systems such as MYCIN and DENDRAL demonstrated the value of structured knowledge representation [26, 27]. Cognitive architectures such as Soar and ACT-R also made internal states and operator applications available for step-level review [28, 29]. In information systems, related work on knowledge management and algorithmic accountability emphasizes that AI artifacts must be represented in ways that support transparency, traceability, and organizational oversight [30, 31].

Modern knowledge-intensive pipelines connect language models with structured knowledge. Knowledge graphs provide a representation substrate for entities, relations, constraints, and quality signals [32]. RAG and GraphRAG expose passages, entity graphs, and community summaries as evidence [33–35]. KG-RAG, knowledge-enhanced language models, and LLM-KG integration expose entity links, triples, and relation paths as reviewable artifacts [36–38]. Recent surveys and KG-quality studies show why graph-quality signals become part of the audit surface [39–41]. Similar pressure appears in long-tail KG-augmented QA and engineering-design RAG, where sparse support and heterogeneous evidence make exhaustive review unrealistic [42–44].

Across these settings, the challenge is no longer simply recovering intermediate evidence. Modern pipelines already expose passages, entity links, relation paths, and verification records. The harder question is how to allocate limited curation budget and preserve records for refinement, update, and reuse [45, 46]. SC-FMA converts exposed artifacts into graph-aware audit records for that purpose.

2.4 Summary of Research Gap

Existing work supplies three ingredients: local attribution identifies salient evidence, process supervision supplies step-level fidelity fields, and graph-based knowledge engineering exposes structural context. The proposed framework combines them into a representation layer that turns exposed knowledge artifacts into inspectable records for fixed-budget knowledge audit.

2.5 Comparison with Alternative Representations

The experimental comparisons below use internal scalar views and reproducible ablations aligned with the audit-record object. IG, SHAP, LIME, GNNExplainer, frozen PRM scorers, and related methods define neighboring objectives whose native outputs include feature attributions, local explanations, explanatory subgraphs, and scalar reward fields. The proposed framework targets a decomposed audit record with fidelity, structural dependency, redundancy, bottleneck, audit-reason, and interpretation fields.

3 Method

3.1 Problem setting

We study fixed-budget audit over directed dependency-flow graphs. Let $G = (V, E)$ be a directed graph whose nodes are auditable or scaffold artifacts and whose edges encode dependency flow. A trace may contain entity bindings, relation paths, retrieved evidence, validation records, or rule-derived intermediate artifacts. The audit budget permits only a subset of nodes to be inspected, so the system must expose which nodes carry structural risk before any human review is claimed.

The main notation is as follows. A *dependency flow* is a directed edge $(u, v) \in E$ indicating that artifact v depends on artifact u for audit interpretation. An *auditable node* $v \in V_A$ is an artifact that can be inspected under the review budget; a *scaffold node* $v \in V_S$ supports graph structure but is not counted as an ordinary audit target. The audit budget is K , and the selected audit set

Table 1 Output objects of neighboring explanation methods and SC-FMA. The table positions each method family by its native representation.

Method family	Native output object	Relation to audit-record construction
IG, SHAP, LIME	Feature-level or local attribution scores	Provides salience signal.
GNNExplainer-style graph explanation	Compact explanatory subgraph	Provides structural subset.
Frozen PRM or process scorer	Scalar step or prefix signal	Provides fidelity field.
SC-FMA	Decomposed audit record	Constructs fidelity, dependency, redundancy, bottleneck, reason, and interpretation fields.

is $S_K \subseteq V$. The scalar raw risk score $r(v)$ is a trace-local normalized impact score used only for same-stratum tie breaking. For selected nodes S_K , Impact Coverage@K is the fraction of unselected auditable descendants in $V_A \setminus S_K$ reached from S_K by directed transitive closure.

The method has two parts. First, a structural label extractor converts graph topology into boolean labels. Second, a Life-Saving First allocation policy uses those labels to select audit candidates under a fixed budget. The method is deliberately not a universal scorer. It does not claim to improve model reasoning, recover causal effects, or validate production knowledge-base maintenance.

3.2 Structural label extractor

The extractor emits two audit-ready boolean fields for each node: **bottleneck** and **redundancy**. A node is marked as a bottleneck when removing it lowers reachable terminal-flow coverage from the frozen source set, or when it lies in a critical source-to-terminal dependency position. This is a graph-theoretic label, not a learned quality prediction.

A node is marked as redundant when its dependency coverage substantially overlaps another node. We define redundancy gold labels using Jaccard > 0.85 over dependency coverage, acknowledging that strict identical coverage is rarely observed in semantic KGs. The corresponding implementation threshold is $\theta = 0.85$; threshold sensitivity at $\theta = 0.90$ is reported in the supplementary scalability analysis.

The TF-IDF comparator is named Semantic-Similarity Baseline (TF-IDF), not a graph-based method. This baseline is included to show that undirected semantic similarity, which lacks flow direction, cannot recover structural bottleneck/redundancy labels.

3.3 Cache locking and reproducibility

The Countries-KG label validation script writes a deterministic cache at `outputs/countries_kg_label_validation/countries_kg_labels_cached.json`. The cache stores trace and node identifiers, `is_bottleneck`, `is_redundant`, redundancy group identifiers, downstream impact counts, auditable flags, thresholds, seed, KG metadata hash, and synthetic graph configuration. The audit case loads this cache and does not recompute graph labels. This cache-locking rule ensures that the label-validation table and the budget-allocation table use the same label source.

All reported Countries-KG and synthetic graph runs use seed 20260711. The semantic fixture contains 30 entities and 189 triples. The synthetic scalability appendix adds directed dependency graphs with 100, 200, 500, 1000, and 5000 nodes, using the 30-node Countries-KG result as the left anchor.

3.4 Life-Saving First allocation policy

Budget allocation proceeds sequentially. For the Countries-KG audit case, the budget is fixed as $K = \lceil 0.25|V_A| \rceil$ selected nodes per trace, where V_A is the auditable-node set; this yields $K = 1$ for 540 repeated traces and $K = 2$ for 60 repeated traces under the locked report. If a layer cumulatively exceeds K , selection stops within that layer sorted by downstream impact; subsequent layers are not

invoked for this trace. The four strata are Critical Bottleneck, Unique Evidence, Redundancy Group Samples, and Fallback.

Critical Bottleneck includes all nodes with `is_bottleneck=True`; these nodes are not excluded for being redundant because a backed-up bottleneck may still require audit visibility. Unique Evidence includes non-redundant high-impact nodes. Redundancy Group Samples allocate at most one representative per redundancy group. Fallback covers remaining candidates only after the earlier strata have not exhausted the budget.

Within a stratum, nodes are ordered by downstream impact and then by the shared raw risk score only to break ties. Raw risk score is used exclusively as a tie-breaker within the same stratification layer, not as the primary allocation driver. Flat Top-K uses the shared raw risk score as the sole ranking criterion. In the KG setting, this scalar is trace-local min-max normalized downstream impact count; deterministic degree and node identifiers break exact ties.

3.5 Impact Coverage@K

Impact Coverage@K counts affected unselected auditable descendants reached from selected nodes under the budget. Formally, descendants are computed by directed transitive closure from each selected node, and selected nodes and non-auditable scaffold/root nodes are excluded from the denominator. Impact Coverage counts all reachable descendants (transitive closure) from selected nodes, not only direct successors. This reflects the maintenance reality that a root bottleneck can affect a long downstream chain, and auditors need visibility into the entire affected subgraph.

The audit report also records Average Path Length to Covered Descendants, Early Truncation Rate, and Budget Used. Average path length is computed from shortest directed distances over the same reachable-descendant set used by Impact Coverage@K. Early truncation records whether the budget is exhausted inside an earlier stratum before later strata are invoked.

3.6 Wikidata controlled maintenance protocol

This experiment validates the proposed audit representation under controlled knowledge-maintenance scenarios on a real KG substrate, rather than evaluating native Wikidata structural roles. The corrected substrate contains scientists born in 1980 or later from physical, life, computing, and social-science occupations. Entity-only relations are extracted to three hops, duplicate predicates are aggregated by ordered node pair, and the cleaned graph is layered into a directed acyclic audit overlay. Layer 3 contains out-neighbors of Layer 2 that do not occur in Layers 0–2.

Evaluation is performed against controlled audit motifs rather than native Wikidata annotations. Each of 30 seeds adds bottleneck, redundant-pair, and degree-matched control records without exposing motif identifiers to the extractor. The primary budget is fixed at $K = 25\%$; $K \in \{5, 10, 15, 20, 25, 30, 40, 50\}\%$ is reported only as sensitivity analysis. Edge deletion and erroneous edge insertion are evaluated separately at 0–20%. For each seed, the absolute selection budget is computed once from the clean-reference candidate count and reused at every noise rate. Selection uses the perturbed graph, while all noise outcomes and structural-protection denominators are scored on the clean reference graph.

The low-budget confirmation partitions each discipline into four stable SHA-256 anchor clusters and uses the 16 clusters, not motif seeds, as paired statistical units. Paired Wilcoxon tests use Holm correction and report Cliff’s delta with bootstrap confidence intervals. Revision-history cases require JSON claim differences between parent and updated revisions. The corrected cache is SHA-256 locked; it supersedes a physical-science-only pilot cache and is not a same-graph rerun.

4 Results

Table 2 shows the proof-of-concept label check. The structural extractor reproduces the fixture-defined labels, whereas the undirected semantic baseline fails to reproduce directed-flow bottleneck and redundancy roles. Betweenness centrality and directed out-closeness are graph-structural bottleneck baselines that select the same number of nodes as the controlled bottleneck reference. The result supports implementation fidelity on a clean semantic fixture; it does not evaluate intrinsic Wikidata roles.

The perfect F1 scores for the structural extractor in Table 2 are expected because the extractor implements the deterministic rules that define the controlled reference labels. Directed out-closeness

Table 2 Countries-KG proof-of-concept and implementation check for structural label F1. The semantic fixture contains 30 entities and 189 triples; labels are cached with seed 20260711. Graph-centrality baselines select the same number of highest-centrality nodes as controlled reference bottlenecks within each trace, so redundancy F1 is not applicable (N/A).

Method	Bottleneck F1	Redundancy F1	Redundancy positives
Structural label extractor	1.000	1.000	124
Semantic-Similarity Baseline (TF-IDF)	0.242	0.405	124
Betweenness Centrality	0.700	N/A	124
Directed Out-Closeness Centrality	1.000	N/A	124

Table 3 Countries-KG Impact Coverage@K at the fixed $K = 25\%$ budget. Flat Top-K uses the shared raw risk score; all methods share the same candidate universe and budget.

Method	Impact Coverage@K	Avg. path length	Early truncation	Budget used
Life-Saving First	1.000	1.000	0.000	1.000
Flat Top-K	0.733	0.828	0.000	1.000
Degree Centrality	0.650	0.761	0.000	1.000
Random Stratified	0.544	0.656	0.000	1.000
Position	0.000	0.000	0.000	1.000
Random	0.333	0.622	0.000	1.000
No-Fallback Ablation	1.000	1.000	0.000	1.000

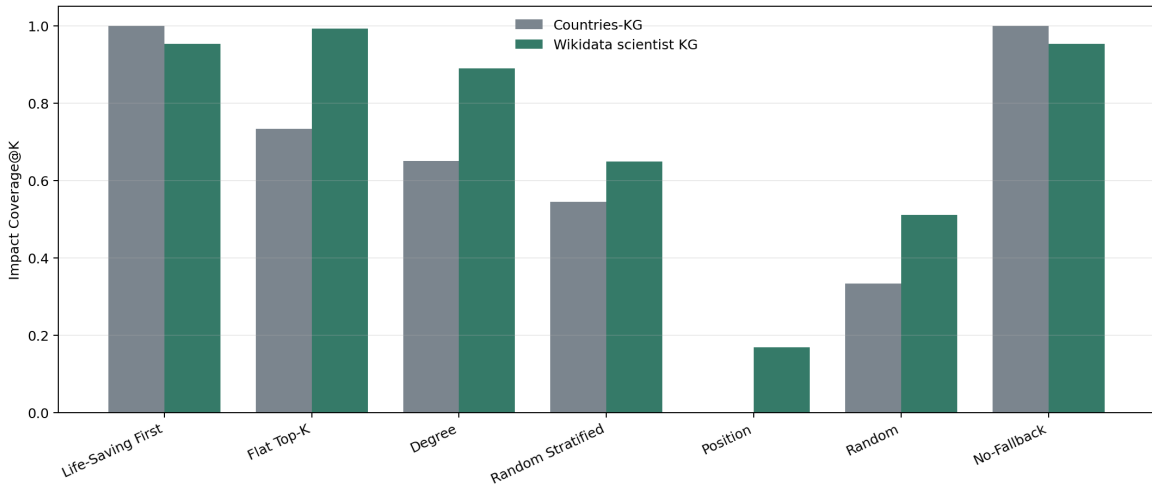


Fig. 2 Countries-KG and Wikidata Impact Coverage at their prespecified $K = 25\%$ budgets. The bars are descriptive within each dataset; no cross-dataset significance test is performed. LSF-Clustered is omitted because no Countries-KG counterpart exists.

also coincides with bottleneck labels in this small fixture, but provides no redundancy field. The table is therefore an implementation-consistency check, not out-of-sample classification evidence.

Table 3 evaluates the cached Countries-KG fixture at $K = 25\%$. Random Stratified shuffles the role fields while preserving staged selection. The No-Fallback Ablation matches the main policy in this fixture, showing internal consistency with the frozen layer ordering; it does not establish that the same ordering will dominate on another graph.

Figure 2 shows that the Countries-KG ordering does not transfer unchanged to the real substrate. This is the intended role of the external experiment: it tests the allocation policy rather than using the clean fixture as evidence of general superiority.

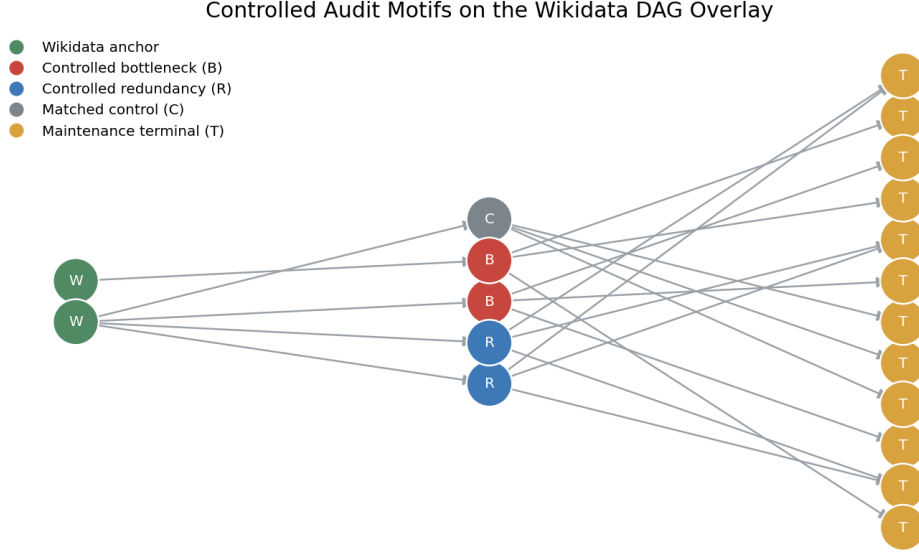


Fig. 3 Controlled audit motifs attached to real Wikidata anchors. The extractor reads topology only; the motif manifest and controlled identifiers are restricted to evaluation.

Table 4 Wikidata primary-budget results over 30 paired motif seeds ($K = 25\%$). Values are mean $\pm$ SD except path length.

Method	Impact Coverage	At-risk protection	Sink-drop mass	Avg. path
Life-Saving First	0.953 ± 0.002	0.972	0.782	1.342
LSF-Clustered	0.952 ± 0.002	0.968	0.785	1.337
Flat Top-K	0.993 ± 0.001	0.895	0.595	1.511
Degree Centrality	0.890 ± 0.002	0.891	0.645	1.361

4.1 Controlled Wikidata audit results

The corrected Wikidata graph contains 2000 nodes, 6039 directed edges, one weakly connected component, diameter 9, and largest-component average shortest-path length 4.189. Its 293 scientist anchors comprise 84 physical-science, 56 life-science, 26 computing, and 127 social-science entities. Controlled Audit Role Evaluation yields bottleneck F1 1.000 and redundancy F1 1.000 over 20 bottleneck, 40 redundant, and 20 matched-control candidates. These values indicate consistency with the controlled motifs, not native Wikidata annotation quality.

Table 4 exposes a coverage–protection trade-off. At $K = 25\%$, Flat Top-K covers more reachable descendants than Life-Saving First (0.993 versus 0.953; paired difference -0.040 , Holm-adjusted $p < 0.001$, Cliff’s $\delta = -1.000$). Life-Saving First instead protects more uniquely at-risk records (0.972 versus 0.895) and greater sink-drop mass (0.782 versus 0.595). It is therefore not a universal Impact Coverage maximizer.

The layer report explains this trade-off. Of 1288 candidates, 537 are classified as Critical Bottleneck and the budget admits 322; all 322 selections occur in the first stratum, while Unique Evidence, Redundancy Group Samples, and Fallback receive zero budget. No-Fallback consequently matches Life-Saving First. LSF-Clustered reduces selected redundancy waste from 0.025 to 0 but does not improve coverage or at-risk protection. The observed policy has collapsed to bottleneck-first selection at the primary budget.

At $K = 5\%$ on the full substrate, Life-Saving First exceeds Flat Top-K by 0.021 (Holm-adjusted $p < 0.001$, Cliff’s $\delta = 0.539$). This seed-level result does not confirm under the stricter anchor-cluster unit. Across 16 prespecified discipline clusters, Life-Saving First is 0.960 ± 0.048 and Flat Top-K is 0.956 ± 0.051 ; the difference is 0.004 with Holm-adjusted $p = 0.385$ and $\delta = 0.051$ (95% bootstrap interval $[-0.020, 0.156]$). LSF-Clustered is identical to Life-Saving First in this confirmation.

Noise is scored on the clean reference graph, so inserted erroneous edges cannot directly inflate evaluation reachability. At 20% deletion, Life-Saving First falls from 0.953 to 0.932 while Flat Top-K

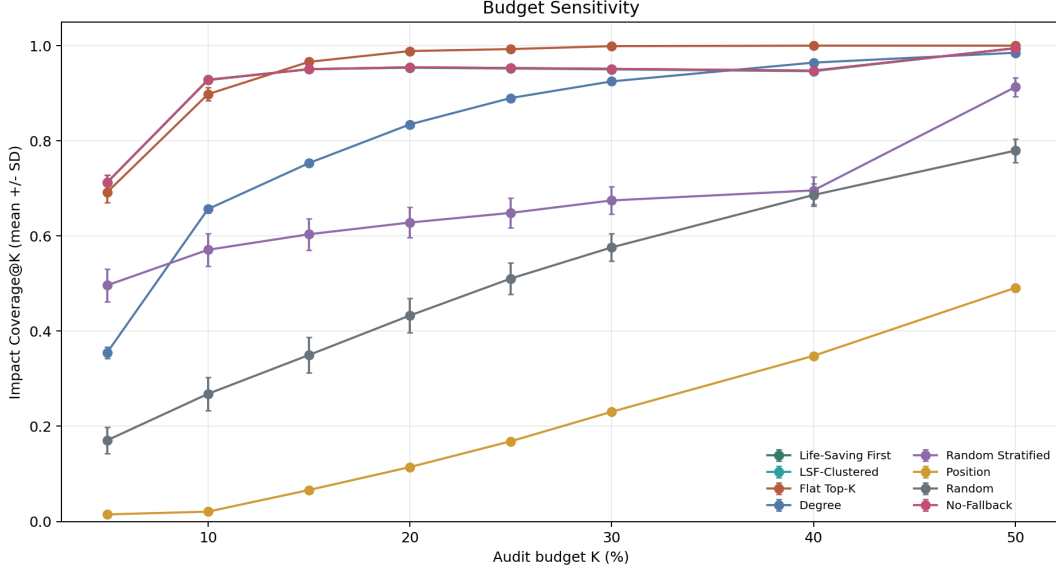


Fig. 4 Wikidata budget sensitivity over 30 paired seeds. Life-Saving First has a small advantage at low budgets, whereas Flat Top-K dominates reachability coverage from $K = 15\%$ onward. Error bars show one standard deviation.

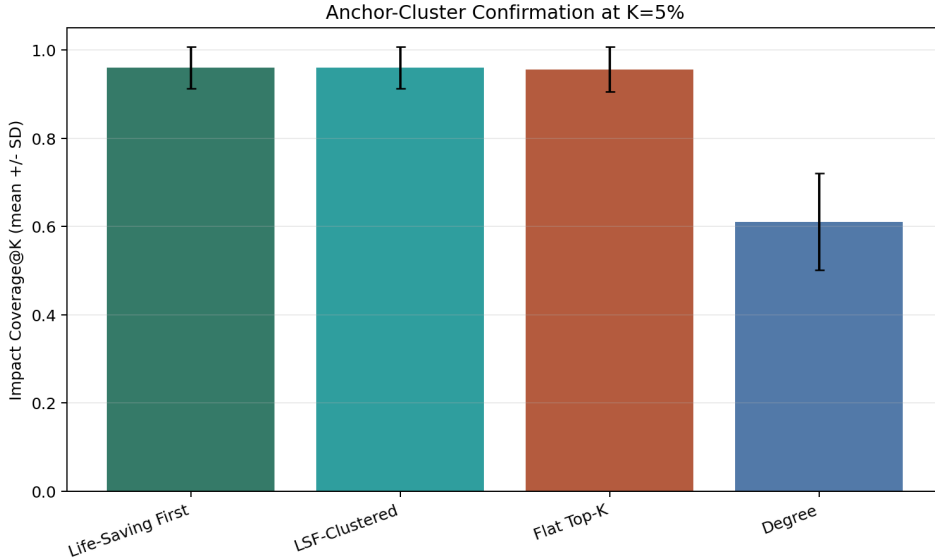


Fig. 5 Prespecified $K = 5\%$ confirmation with 16 discipline anchor clusters as paired units. The small Life-Saving First–Flat Top-K difference is not statistically significant after Holm correction.

falls from 0.993 to 0.987. At 20% insertion, Life-Saving First falls to 0.878 while Flat Top-K rises descriptively to 0.996 because noisy selection changes which nodes are chosen, not because erroneous edges are counted during scoring. Relative to Flat Top-K, Life-Saving First degrades more for Impact Coverage and at-risk protection under both perturbations (Holm-adjusted $p < 0.001$). Table 5 reports the endpoints.

Efficiency measurements use one warm-up and ten measured runs per size. Mean total runtime is 0.643, 0.714, and 5.855 seconds at 500, 1000, and 2000 nodes; peak traced Python memory is 2.175, 3.893, and 8.769 MB. The empirical log–log runtime slope is 1.594. Role extraction dominates at 2000 nodes (5.570 seconds), consistent with the pairwise coverage-comparison stage; these observations document scaling rather than theoretical optimality.

Finally, two verified Wikidata revision events ground the maintenance interpretation: a P108 institution addition for Q100156655 (revision 1288194332) and a P166 award addition for Q100166319 (revision 1348244988). JSON claim differences confirm the old and new values. In the controlled one-of-four overlay, Life-Saving First selects the sole bottleneck and protects three uniquely at-risk

Table 5 Clean-reference noise endpoints over 30 paired seeds. IC is Impact Coverage and AR is protected at-risk coverage.

Noise	Method	IC@0	IC@20%	AR@0	AR@20%
Deletion	Life-Saving First	0.953	0.932	0.972	0.940
Deletion	Flat Top-K	0.993	0.987	0.895	0.895
Insertion	Life-Saving First	0.953	0.878	0.972	0.900
Insertion	Flat Top-K	0.993	0.996	0.895	0.892

records, while Flat Top-K selects a bypassed high-impact control. The revision events are real; the prioritization scenario is controlled and was not part of the historical editing decision.

5 Appendix A. Synthetic Scalability and Threshold Sensitivity

Countries-KG is deliberately used as a clean semantic fixture, not as a scale claim. To avoid making the 30-node fixture carry the full generalization burden, we add a seeded synthetic scalability test with directed dependency graphs of 100, 200, 500, 1000, and 5000 nodes. The generator plants source-to-terminal chains, bottleneck positions, and redundant dependency-coverage twins so that the graph-theoretic truth is known before extraction. This makes the test an implementation-scalability and controlled-label-recovery check rather than a noisy open-domain KG validation.

The synthetic graphs use seed 20260711 and the same redundancy threshold family as the semantic fixture. Table 6 places Countries-KG at the left anchor and then reports macro F1 across the synthetic scale points. The structural extractor remains at macro F1 1.000 through 5000 nodes because the synthetic truth is generated from the same deterministic bottleneck and redundancy rules. This is expected: the point is to check whether the implementation preserves rule fidelity as graph size grows, not to report a learned classifier’s out-of-sample accuracy.

Table 6 Appendix A synthetic scalability curve. Countries-KG is shown as the 30-node semantic left anchor; the remaining rows are seeded directed dependency graphs.

Source	Nodes	Bottleneck F1	Redundancy F1	Macro F1
Countries-KG	30	1.000	1.000	1.000
Synthetic directed graph	100	1.000	1.000	1.000
Synthetic directed graph	200	1.000	1.000	1.000
Synthetic directed graph	500	1.000	1.000	1.000
Synthetic directed graph	1000	1.000	1.000	1.000
Synthetic directed graph	5000	1.000	1.000	1.000

The threshold-sensitivity check compares $\theta = 0.85$ with $\theta = 0.90$ for redundancy coverage. In this planted fixture, both thresholds recover the same macro F1 across the left anchor and all synthetic scale points. This does not imply that arbitrary noisy KGs are threshold-insensitive; it only shows that the implementation is stable under the two planned thresholds when the redundancy relation is known.

Table 7 Redundancy-threshold sensitivity for the Appendix A scalability check. Values are macro F1.

Threshold	30	100	200	500	1000	5000
$\theta = 0.85$	1.000	1.000	1.000	1.000	1.000	1.000
$\theta = 0.90$	1.000	1.000	1.000	1.000	1.000	1.000

The path-length diagnostic prevents Impact Coverage from being misread as direct-neighbor counting. Synthetic covered descendants have average path lengths above one and increase with graph

size, from 2.587 at 100 nodes to 3.883 at 5000 nodes. Thus, the coverage implementation traverses reachable descendants by directed transitive closure rather than stopping at immediate successors.

Table 8 Average path length to covered descendants in the seeded synthetic scalability check.

Nodes	100	200	500	1000	5000
Average path length	2.587	2.819	3.115	3.349	3.883

The scalability appendix is paired with a cache-locking checklist because the two main tables depend on the same structural labels. Table 9 records the fields that must remain fixed between label validation and budget allocation. This prevents a common reproducibility error: one script reports label F1 using one graph-construction threshold, while a later audit script silently recomputes labels under a different threshold or seed. In the present package, the audit script loads the cached label file and records a recomputed-label count of zero.

Table 9 Cache-locking checklist for Countries-KG label validation and Impact Coverage evaluation.

Cached field	Reproducibility role
Trace and node identifiers	Keeps Table 2 and Table 3 on the same auditable node universe.
Boolean labels	Stores <code>is_bottleneck</code> and <code>is_redundant</code> rather than letting the audit script infer them again.
Redundancy group identifier	Ensures that the Redundancy Group Samples layer draws representatives from the same group assignment used in validation.
Downstream impact count	Supplies the impact order used inside strata and the shared scalar source for Flat Top-K.
Auditable flag	Excludes scaffold/root nodes from ordinary baseline denominators while still allowing critical bottlenecks to enter the first stratum.
Thresholds and seed	Records $\theta = 0.85$, the alternative $\theta = 0.90$ sensitivity check, and seed 20260711.
KG metadata hash	Confirms that the 30-entity, 189-triple semantic fixture has not changed between validation and audit allocation.

This checklist is deliberately operational. It does not expand the empirical claim beyond the clean semantic fixture. It only states that the reported F1 comparison and Impact Coverage simulation are label-consistent, seed-locked, and derived from the same cached dependency-flow records.

6 Boundary Diagnosis

Countries-KG supplies an internal consistency check under a directed dependency fixture, but the representation should not be assumed informative when graph semantics are weak. We therefore retain a compact exploratory boundary check on PRM800K, where the graph constructor supplies sparse topical edges rather than explicit dependency flows. Appendix B reports this route as exploratory boundary evidence, not as a necessary-condition or causal test.

In this route, TF-IDF graph edges are sparse and directionless relative to the explicit dependency semantics required by the KG audit representation.

The compact finding is that annotation-order consistency on PRM800K is dominated by a supervised fidelity field and positional/lexical cues. The supervised w_{struct} field reaches Spearman $\rho = 0.611$ on the locked split. A same-supervision structure-only Ridge control reaches only $\rho = 0.043$, while adding position raises it to $\rho = 0.603$. Thus, graph-derived fields do not provide independent annotation-order signal in this route; they remain organizational audit-record fields rather than validated ranking predictors.

Table 10 Exploratory PRM800K boundary diagnosis. The table is a compact main-text pointer to Supplementary Appendix B and is not used as core evidence for the KG audit-allocation claim.

Field or control	Spearman ρ	Interpretation
w_{struct} (supervised fidelity field)	0.611	Strong process-annotation signal
Structure graph + position	0.603	Mostly position-associated
Structure graph only	0.043	Weak independent graph signal

This negative boundary supports the paper’s scope. Structural labels require directed dependency flows with interpretable graph semantics; they should not be expected to transfer unchanged to noisy process-annotation traces whose graph edges are sparse, topical, or directionless. The detailed PRM800K confidence intervals, fixed-budget readouts, and archived scorer-route diagnostics are retained in the supplementary material.

7 Discussion

The central shift is from scalar audit priority to an explicit audit-record representation: reusable graph labels, strata, downstream visibility, and structural-protection fields. Countries-KG remains an implementation fixture. The corrected Wikidata substrate adds a real graph topology, multidisciplinary entities, noise perturbations, revision evidence, and engineering measurements, while controlled motifs keep the evaluation target observable.

Life-Saving First should be read as an illustrative consumer of the representation, not as the paper’s universal optimizer. At $K = 25\%$, it sacrifices reachability coverage for higher protection of uniquely at-risk records and exhausts its budget inside the bottleneck stratum. At $K = 5\%$, its seed-level advantage does not confirm across anchor clusters. The clustered variant removes selected redundancy waste but does not improve the confirmation result. These findings motivate exposing both reachability and structural protection rather than collapsing audit allocation into one headline metric.

The impact-coverage metric should be read as a dependency-flow visibility measure. It asks how much unselected downstream audit surface is affected by the selected nodes under a fixed budget. It does not measure human review speed, decision confidence, correctness of human judgments, production maintenance cost, or causal effect.

8 Limitations

The Wikidata experiment remains controlled. Its F1 target comes from audit motifs attached to a real substrate, not from native Wikidata role annotations. The corrected multidisciplinary cache differs from the earlier physical-science-only pilot substrate, and the provenance record explicitly marks it as not a same-graph rerun. Sixteen anchor clusters are units from one extracted KG, not 16 independent KGs. Edge perturbations test selection robustness against a clean reference graph; they do not reproduce all ontology, temporal, or adversarial errors in production knowledge bases. These experiments do not show that the representation is robust to arbitrary KG noise. Robustness to arbitrary KG noise is not a claim of the current framework.

The study also does not report human usefulness evidence. The submission package contains a blank future human-evaluation protocol, but no returned ratings, evaluator provenance, declarations, timestamps, hashes, or reliability analysis are available. The present evidence therefore concerns structural label extraction and Impact Coverage on KG dependency flows, not human decision time or human audit accuracy. It is not production knowledge-base validation.

Future work should therefore use independently extracted KGs and native maintenance outcomes, compare multi-objective allocation policies that explicitly trade reachability against at-risk protection, and conduct provenance-complete user studies of auditor efficiency, accuracy, and prioritization. Learned role predictors are relevant only if their targets are independently annotated rather than regenerated from the extractor’s own rules.

9 Conclusions

This paper reframes SC-FMA as an audit-record representation for fixed-budget knowledge maintenance. The graph-topology interface emits bottleneck, redundancy, downstream-impact, and protection fields that can support multiple allocation policies. Countries-KG and seeded synthetic graphs establish implementation consistency; the corrected Wikidata substrate provides a controlled real-KG test, noise diagnostics, anchor-cluster confirmation, efficiency measurements, and verified revision context. The results do not support Life-Saving First as a universal coverage maximizer: its structural-protection advantage at $K = 25\%$ accompanies lower reachability coverage, and its low-budget advantage over Flat Top-K is not significant across anchor clusters. The contribution is therefore the inspectable representation and evidence boundary, not production validation, human usefulness, evaluation of intrinsic Wikidata roles, or causal identification.

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Declaration of Competing Interest

The authors declare that they have no conflicts of interest related to this work.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used OpenAI GPT-5 to improve language clarity and assist with LaTeX formatting checks. The tool was not used to design the study, conduct data analysis, generate results, or derive the conclusions. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Data Availability

The PRM800K process-annotation dataset, MuSiQue, WebQSP, and Wikidata revision histories are publicly available from their original sources. The locked Wikidata cache hash, SPARQL query log, triples, motif manifests, seed-level metrics, verified revision differences, figures, configurations, and reproduction scripts are included in the experiment outputs and will be released through the review repository where permitted.

CRedit authorship contribution statement

Haoran Ma: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data Curation, Writing – Original Draft, Visualization. **Ningning Wang:** Methodology, Validation, Writing – Review & Editing, Supervision, Project administration, Funding acquisition.

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